A Survey of LLM-Based Question Answering Architectures and the Role of Semantics in Enterprise Decision Support Systems

Antony Seabra[a] Daniel Schwabe[b] and Sergio Lifschitz[c]
Departamento de Informatica, PUC-Rio, Brazil
{*amedeiros@inf.puc-rio.br, dschwabe@gmail.com, sergio@inf.puc-rio.br*

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Abstract: Decision support systems (DSS) are central to enterprise environments, where complex decisions require integrating heterogeneous information, applying domain criteria, and interpreting implicit knowledge. Recent advances in large language models (LLMs) offer new opportunities to support such processes through natural language interaction and flexible access to distributed data. However, current LLM-based question answering architectures remain limited in their ability to address decision-oriented queries, as they largely depend on retrievable information and lack explicit representations of domain semantics. This survey analyzes contemporary approaches—including retrieval-based, graph-based, and agentic architectures—with respect to their representational assumptions, reasoning mechanisms, and suitability for decision support. We identify a shared structural constraint across these paradigms: their reliance on accessible data rather than formalized conceptual knowledge. To address this limitation, we examine how semantic representations and structured knowledge models, such as ontologies and symbolic reasoning frameworks, can complement existing architectures. We conclude by outlining research directions for developing semantically grounded AI systems capable of supporting complex decision-making in enterprise contexts.

1 INTRODUCTION

Decision-making in enterprise information systems increasingly depends on the ability to formulate, interpret, and answer complex questions. In domains such as compliance, risk management, procurement, and governance, decision-makers must integrate heterogeneous information, evaluate constraints, and reason about incomplete or distributed knowledge. In such environments, obtaining relevant information is only one component of the decision process; equally important are interpretation, abstraction, and contextualization. These requirements place growing demands on computational systems intended to support decision-making, which must not only retrieve data but also assist in transforming information into actionable understanding.

Large language models (LLMs) are promising for decision support because they combine broad linguistic competence with the ability to synthesize information across heterogeneous sources. Trained on large corpora, they encode extensive statistical knowledge that enables users to explore information spaces, formulate queries, and interpret results. In enterprise settings, this reduces cognitive load, accelerates analysis, and supports iterative decision-making by allowing interaction with complex data ecosystems through natural language, positioning LLMs as interfaces for accessing and integrating organizational knowledge.

However, current LLM-based question-answering architectures operate largely over surface-level representations such as embeddings, similarity search, and structured queries. While effective for retrieval and summarization, they lack explicit semantic representations of the concepts, relationships, and assumptions underlying decision processes. This limitation is evident in scenarios involving abstract notions like risk, criticality, or compliance, which are not directly encoded in data sources. As a result, systems may produce plausible but semantically misaligned answers, reflecting a structural reliance on data availability rather than explicit semantic modeling, a limitation that persists even in recent extensions such as graph-based retrieval, tool-augmented reason-

[a] https://orcid.org/0009-0007-9459-8216
[b] https://orcid.org/0000-0003-2825-3352
[c] https://orcid.org/0000-0003-3073-3734

ing, and agentic approaches.

This paper presents a survey of LLM-based question-answering architectures, analyzing their representational foundations, reasoning mechanisms, and applicability to decision-support tasks. Examining paradigms such as retrieval-based, graph-based, and agent-oriented approaches, it identifies structural characteristics that shape their strengths and limitations, and positions the discussion at the intersection of decision support, question answering, retrieval, grounding, and semantic representation. The survey contributes by providing a structured taxonomy of current architectures, a comparative analysis highlighting their shared dependence on explicitly available data, and research directions for incorporating explicit semantic components to better support decision-making tasks that require conceptual interpretation rather than literal retrieval.

The remainder of this paper is organized as follows. Section 2 presents the conceptual background. Section 3 introduces a taxonomy of LLM-based question-answering architectures. Section 4 provides a comparative analysis. Section 5 discusses semantic-oriented extensions. Finally, Section 6 concludes and outlines future research directions.

2 BACKGROUND

Question Answering (QA) systems map natural language questions to answers derived from one or more data sources and, in enterprise environments, act as interfaces to heterogeneous repositories such as documents, databases, and analytical systems. In decision-support scenarios, however, correctness alone is insufficient, as answers must support interpretation, justification, and action. This aligns with classical decision science, which emphasizes that decision making relies on abstractions and reasoning rather than raw data alone (Simon, 1960), and with decision support systems research that distinguishes data-driven retrieval from model-driven reasoning requiring explicit representations of domain concepts (Power, 2002). Accordingly, a distinction emerges between fact-oriented questions, answerable through explicit evidence, and decision-oriented questions, which depend on conceptual interpretation and derived criteria. This reflects a broader methodological divide in QA between tasks solvable through direct evidence access and those requiring integration of distributed information, as observed in complex QA benchmarks. In enterprise contexts, where information is fragmented across diverse systems, the effectiveness of QA systems therefore depends on their ability to access, retrieve, and organize external knowledge, motivating architectures that extend language models with mechanisms to interact with information beyond their parametric memory.

A key driver of LLM adoption in enterprise QA is their ability to access knowledge beyond model parameters. Retrieval-augmented generation (RAG) achieves this by coupling an LLM with an external corpus through indexing, retrieving top-k items, and augmenting prompts with retrieved content (Lewis et al., 2020). This expands information access and freshness, enabling models to use up-to-date enterprise data. Nevertheless, RAG introduces design sensitivities, leading to variants that differ in query formulation, evidence selection and structuring, and mechanisms for handling retrieval errors.

One line of work improves retrieval by refining query representations or organizing evidence more effectively. HyDE generates hypothetical documents to guide embedding-based retrieval, while RAPTOR structures corpora into hierarchical summaries to better capture dispersed context (Gao et al., 2023a). A complementary direction addresses limitations of retrieval itself in enterprise settings, where it can be noisy or unnecessary. Approaches such as Self-RAG enable models to decide when retrieval is needed and to assess the support of generated answers, while CRAG introduces evaluators that trigger corrective retrieval actions when initial results are unreliable (Asai et al., 2024). Together, these methods enhance retrieval quality and control through query reformulation, hierarchical organization, and feedback mechanisms, but remain grounded in textual evidence rather than explicit semantic representations.

Graph-based retrieval offers an alternative when questions require global sensemaking across a corpus, where standard RAG struggles due to dispersed information. Approaches like GraphRAG construct entity-centric graphs and retrieve aggregated community-level summaries, shifting the focus from isolated passages to structured representations of cross-document relationships (Edge et al., 2024). In enterprise settings, external knowledge access often also involves structured systems such as relational databases, where Text-to-SQL translates natural language into executable queries, grounding answers in schema elements and constraints rather than textual similarity (Gao et al., 2024). More broadly, tool-augmented paradigms extend this idea by enabling LLMs to invoke external APIs and systems during reasoning (Schick et al., 2023), emphasizing mechanisms for accessing, filtering, and structuring external information rather than explicit semantic modeling.

Taken together, retrieval- and tool-augmented ar-

chitectures expand the operational scope of LLM-based QA systems by enabling access to private corpora, structured data, and aggregated knowledge representations (Gao et al., 2023b). Nonetheless, they largely assume that decision-relevant meaning can be recovered from retrieved text, summaries, or query results, which motivates a shift toward viewing LLMs not only as generators conditioned on evidence, but as interfaces that structure and coordinate multi-step reasoning over it.

2.1 LLMs as Reasoning Interfaces

In contemporary QA pipelines, LLMs are used not only as answer generators, but also as semantic parsers, decomposition engines, and interactive reasoning components that orchestrate multi-step information access. Retrieval-augmented approaches combine language models with external sources to improve factual grounding (Lewis et al., 2020), while dense retrieval methods enable efficient semantic matching between queries and passages (Karpukhin et al., 2020). However, although these mechanisms expand access to external knowledge, they do not ensure a grounded understanding of the underlying concepts. Semantic grounding refers to the alignment between model outputs and real-world meanings of entities and relations, rather than mere statistical associations (Bender et al., 2021), and prior work shows that even advanced models can produce fluent but semantically ungrounded responses in complex reasoning tasks.

Recent research therefore emphasizes the role of intermediate reasoning structures that help models connect concepts and evidence more coherently. Techniques such as chain-of-thought prompting expose intermediate steps and improve performance on multi-step tasks (Wei et al., 2022), yet they remain vulnerable when the semantics of the problem are not properly represented. In such cases, reasoning traces may appear plausible while lacking conceptual validity. This issue is compounded by findings on factuality and trustworthiness, which show that reasoning over weakly grounded content can propagate errors and produce confident but incorrect answers.

These limitations are particularly critical in enterprise decision-support scenarios, where interpretability and semantic validity are essential. Consequently, effective reasoning in LLM-based QA systems depends on strong semantic grounding, ensuring that outputs are aligned with meaningful and conceptually coherent evidence rather than surface-level patterns.

2.2 Knowledge Representations

A central design dimension in intelligent systems concerns how knowledge is represented for reasoning. Knowledge representation research distinguishes between symbolic representations, which explicitly encode concepts, relations, and constraints, and distributed representations, which capture statistical associations in vector spaces (Brachman and Levesque, 2004). These paradigms provide complementary capabilities: distributed representations enable similarity-based retrieval and generalization, while symbolic representations support interpretation, constraint validation, and concept-level reasoning. In decision-support contexts, this distinction is particularly relevant because many decision-relevant constructs are not directly stored as data but must be derived from domain definitions, policies, or criteria.

Such constructs highlight the need for explicit conceptual grounding beyond textual similarity. Classifications such as risk level, compliance status, or contractual criticality often depend on multiple attributes combined according to predefined rules or definitions, and therefore cannot be reliably inferred from surface-level evidence alone. Systems designed for decision support must therefore incorporate mechanisms that link natural language inputs to structured conceptual knowledge, enabling the interpretation of derived or abstract notions that are not explicitly encoded in underlying data sources.

Semantic technologies provide formal mechanisms to address this challenge by representing domain meaning through structured models. Ontologies, defined as formal specifications of shared conceptualizations (Gruber, 1993), enable machine-interpretable representations of entities, relations, and constraints, following principles such as clarity and coherence (Uschold and Gruninger, 1996). Complementary representations, including knowledge graphs, logical rule systems, and schema-based models, further support integration and reasoning across heterogeneous sources (Berners-Lee et al., 2001). Within LLM-based architectures, these representations act as semantic substrates that ground language understanding in explicit conceptual structures, enabling hybrid reasoning processes that combine linguistic interpretation with conceptually grounded decision support.

3 Taxonomy of LLM-Based Question Answering Architectures

This section proposes a taxonomy of contemporary LLM-based question answering (QA) architectures, focusing on how they access knowledge, structure evidence, and operationalize multi-step reasoning and tool use. Rather than organizing approaches by technique alone, the taxonomy is grounded in architectural commitments such as representation, retrieval organization, orchestration, and feedback, enabling a unified perspective across paradigms. While prior surveys have classified specific subdomains—such as RAG pipelines (Gao et al., 2023b), graph-based retrieval (Peng et al., 2024), and agentic systems (Plaat et al., 2025)—they typically remain confined to individual approaches, lacking a cross-paradigm analytical view. In contrast, the taxonomy proposed here integrates retrieval-based, graph-based, structured, tool-augmented, and agentic architectures into a single framework, emphasizing how knowledge is externalized, structured, accessed, and operationalized for decision-oriented queries. Architectures are classified along four orthogonal axes: (i) externalization of knowledge, (ii) organization of evidence, (iii) control and orchestration, and (iv) feedback and correction. This perspective supports systematic comparison across approaches and highlights structural commonalities that are not evident when architectures are analyzed in isolation.

Flat RAG Pipelines and Their Extensions A large portion of enterprise QA systems follows the retrieval-then-generate paradigm, where a retriever selects top-k passages and an LLM generates answers conditioned on them, with recent work introducing modular improvements such as enhanced retrievers, rerankers, and prompt strategies (Gao et al., 2023b). While effective for fact-based queries, this approach assumes that answers can be recovered from retrieved text at a given granularity, which limits its applicability in more complex scenarios. To address this, recent methods incorporate feedback loops to improve robustness and control: Self-RAG enables models to decide when retrieval is needed and to critique their own outputs (Asai et al., 2024), while CRAG evaluates retrieved evidence and triggers corrective actions when it is unreliable. Despite these advances, such approaches remain fundamentally grounded in textual evidence, highlighting a key limitation for decision-support tasks that require deeper semantic interpretation.

Hierarchical and Multi-Retrieval Architectures A second family modifies the organization of evidence rather than only retriever/generator components. The core motivation is that long documents and complex corpora often require retrieval at multiple abstraction levels, not only at the chunk level. RAPTOR constructs a tree of recursively clustered and summarized text units, enabling retrieval over both fine-grained passages and higher-level summaries (Sarthi et al., 2024). Such multi-resolution retrieval improves holistic coverage and multi-step QA performance in long-context settings. Conceptually, these methods move from flat evidence to structured evidence, while still operating primarily over textual artifacts.

Query Expansion and Synthetic Retrieval Signals Another line of work improves retrieval by generating intermediate textual signals that better capture relevance patterns. HyDE proposes generating a hypothetical document for a given query and embedding it to retrieve real documents in a zero-shot dense retrieval setting (Gao et al., 2023a). This family is useful to frame because it explicitly separates (i) semantic intent capture (via LLM-generated pseudo-documents) from (ii) grounding (retrieval over real corpora). In practice, it can increase recall for under-specified questions, though it still inherits limitations when the required answer depends on non-textual domain definitions or normative criteria.

Graph-Augmented Retrieval and Generation GraphRAG denotes architectures that incorporate graphs as a first-class organizing structure for retrieval and generation. Recent surveys formalize GraphRAG pipelines around graph construction/indexing, graph-guided retrieval, and graph-enhanced generation, and they highlight design choices across domains and graph types (knowledge graphs, entity graphs, document graphs, community graphs) (Peng et al., 2024). Compared to flat RAG, GraphRAG systems can better support multi-hop queries by retrieving relational neighborhoods, communities, or paths, and by using graph structure to constrain evidence aggregation. Importantly, this family introduces an intermediate representation that is more amenable to semantic constraints and relational reasoning, making it a key bridge toward semantic-oriented artifacts discussed later.

Structured Grounding A related family grounds QA in structured stores such as knowledge graphs (KGs) and databases, often via text-to-query translation, for example NL-to-SPARQL, or hybrid graph-and-text retrieval. Recent synthesis work surveys how

Architecture Family	Knowledge Externalization	Evidence Organization	Control Orchestration	Feedback Mechanisms
Flat RAG	External text corpus	Flat passages/chunks	Single or iterative prompting	Optional reranking or retrieval refresh
Hierarchical RAG	External text corpus	Hierarchical summaries + passages	Multi-level retrieval	Limited iterative refinement
GraphRAG	External corpus + Graph index	Relational / graph structures	Graph-guided retrieval	Optional traversal refinement
KG/DB-grounded	structured knowledge bases	schema + relations	query execution + hybrid reasoning	constraint validation
Tool-augmented systems	APIs, tools, modules	tool outputs	tool selection + routing	execution verification
Agentic architectures	tools + memory + retrieval	dynamic working context	planning + iteration	self-reflection / monitoring
Deliberative inference (ToT/PoT)	internal reasoning traces	candidate thoughts/programs	search / execution	branch evaluation

Table 1: Taxonomy-aligned comparison of LLM-based QA architectures along structural design dimensions.

LLMs and KGs can be combined for QA, emphasizing KGs as (i) a source of explicit relations, (ii) a scaffold for multi-hop reasoning, and (iii) an anchor for verifiable retrieval and constraint checking (Ma et al., 2025). This family differs from GraphRAG in that graphs are not merely an auxiliary retrieval structure but an explicit knowledge representation substrate with schema, semantics, and constraints. As such, KG/DB-grounded is central when moving from "answerable from text" to "answerable from conceptual models."

Tool-Augmented and Modular Architectures Tool-augmented QA systems extend beyond retrieval by allowing an LLM to call external APIs and specialized modules (search, calculators, databases, domain engines). Toolformer demonstrates that models can be trained to decide when and how to call tools using self-supervision and a small set of demonstrations (Schick et al., 2023). MRKL systems propose a modular neuro-symbolic architecture in which an LLM routes requests to discrete knowledge and reasoning modules, aiming to separate linguistic processing from specialized capabilities. This family is particularly relevant for enterprise settings because decision support often requires access to structured computation (e.g., pricing analytics) and policy logic (e.g., compliance checks) that are better handled by explicit modules than by pure generation.

Agentic Architectures Agentic QA architectures emphasize control and orchestration: the system decomposes tasks, selects tools, maintains state, and iterates through intermediate steps. ReWOO proposes decoupling reasoning from observations by producing a plan before tool calls, reducing prompt redundancy and enabling more efficient multi-step workflows. More broadly, recent surveys define agentic LLMs as systems that reason, act, and interact, organizing literature around planning, tool use, and feedback learning (Plaat et al., 2025). In enterprise QA, agentic architectures are often used to integrate heterogeneous sources and workflows; however, their success still depends on what semantic criteria are explicitly represented and what remains implicit, motivating semantic augmentation as a complementary direction.

Deliberative Inference A family of methods enhances inference-time reasoning by introducing explicit search and execution mechanisms. Tree-of-Thoughts (ToT) extends chain-of-thought by exploring multiple reasoning paths with lookahead and backtracking (Yao et al., 2023), while Program-of-Thoughts (PoT) delegates computation to external interpreters through executable code generation. Although not retrieval architectures, these approaches enable more controllable and auditable reasoning processes in QA systems. More broadly, recent advances show a trend toward externalizing knowledge, structuring evidence, and improving control through feedback, tools, and agents. Despite this, most systems still assume that decision-relevant meaning can be recovered from data or approximated through language patterns. This survey argues that future progress will depend on integrating deeper semantic representations that connect language, data, and concepts, shifting QA systems from information access toward conceptual interpretation and decision support through advances in semantic modeling, hybrid reasoning, and evaluation.

4 Comparative Analysis

Building on the taxonomy in Section 3, this section compares architectural families based on their design assumptions and their ability to support complex reasoning and decision-oriented queries, rather than benchmark performance. This structural perspective highlights trade-offs, complementarities, and shared limitations across paradigms, while situating recent research within a landscape marked by rapid diversification rather than convergence. Recent work has increasingly evaluated agentic and tool-augmented LLM systems in realistic, interactive settings, with benchmarks focusing on general assistant capabilities, tool use, and long-horizon reasoning challenges, revealing both the potential and current limitations of these architectures.

A central trend across recent architectures is the progressive externalization of knowledge, shifting LLMs from parametric systems to interfaces that dynamically access documents, databases, graphs, and tools. This reframes question answering as the coordination of distributed knowledge sources rather than simple recall, driving innovation toward improved retrieval, evidence integration, and access mechanisms. In parallel, architectures increasingly focus on structuring evidence, moving beyond flat retrieval toward hierarchical and relational representations that allow reasoning over summaries, communities, or graph structures, better supporting queries that require global context.

Another key direction concerns control and orchestration, with agentic and planning-based systems enabling task decomposition, tool selection, and iterative reasoning, positioning LLMs as coordinators of multi-step processes rather than static generators. At the same time, there is growing interest in integrating symbolic and formal reasoning with neural models through hybrid approaches that combine probabilistic inference with structured representations such as knowledge graphs, rules, or ontologies, aiming to improve interpretability, consistency, and conceptual correctness in decision-support scenarios.

Together, these frontiers indicate that the field is transitioning from optimizing isolated components toward designing integrated systems that jointly address access, structure, reasoning, and semantics. Understanding how these dimensions interact is essential for assessing the capabilities and limitations of current architectures, which motivates a comparative analysis between them. While Table 1 lists both architectural paradigms and internal mechanisms (such as retrieval refinement or feedback loops), Table 2 abstracts from these internal variations to compare architectures at a uniform structural level. We evaluate architectures according to five analytical dimensions: knowledge representation substrate, grounding and evidence access, reasoning capability, and decision-support suitability. These dimensions capture structural properties that influence system behavior independently of model scale or training data.

Strengths Across Architecture Families Different architectural paradigms exhibit distinct strengths depending on how they organize evidence and reasoning. Retrieval-based systems excel at factual grounding by conditioning generation on external passages, improving accuracy and reducing hallucinations (Shuster et al., 2021). Hierarchical retrieval further enhances performance in long-context scenarios by structuring information across abstraction levels, enabling integration of dispersed evidence (Taylor et al., 2022). Graph-based approaches add advantages for multi-hop reasoning by explicitly modeling relationships between entities, improving inference across interconnected facts (Sun et al., 2019).

Tool-augmented and modular architectures extend capabilities by allowing LLMs to interact with external systems, such as databases or computational modules, enabling operations beyond text generation (Seabra et al., 2025). Agentic systems build on this by supporting task decomposition, planning, and iterative interaction, making them effective for complex, multi-step problems. Additionally, search-based inference methods improve reasoning reliability by exploring multiple candidate paths rather than a single trajectory, enhancing performance in tasks requiring logical consistency.

Structural Limitations Across Paradigms Despite their differences, existing architectures share structural constraints that limit decision-oriented reasoning. Most assume that relevant knowledge can be retrieved from available data, meaning they struggle when answers depend on implicit domain definitions or conceptual criteria not explicitly encoded in repositories. Additionally, many rely on distributed representations, which capture statistical patterns but not explicit constraints, making them less effective for tasks requiring rule-based or symbolic reasoning.

These limitations are compounded by variability in reasoning transparency and robustness to ambiguous queries. Some systems provide intermediate reasoning traces, while others output only final answers, hindering verification and trust in high-stakes contexts (Lipton, 2018). Moreover, approaches based on similarity-driven retrieval may return plausible but irrelevant results when queries lack context, with iter-

Architecture Family	Representation	Grounding	Reasoning	Decision Support
Flat RAG	text chunks	strong	limited multi-hop	limited
Hierarchical RAG	hierarchical text	strong	improved synthesis	moderate
GraphRAG	graph + text	strong relational	multi-hop	moderate–high
KG/DB-grounded	structured symbolic	explicit	rule/logic capable	high
Tool-augmented	hybrid	explicit	procedural	high
Agentic systems	hybrid + tools	dynamic	multi-step planning	high
Search-based inference	generated traces	internal	strong reasoning	moderate

Table 2: Comparative analysis of architecture families across structural dimensions.

ative or agentic methods offering only partial mitigation depending on the availability of effective refinement signals.

By contrary, conceptual grounding requires representations that explicitly define domain entities, relationships, constraints, and classification rules. Such representations provide a stable semantic reference layer that complements probabilistic inference. Research in hybrid AI systems has demonstrated that combining symbolic reasoning with neural models can improve robustness, generalization, and interpretability, particularly in tasks involving structured knowledge or formal reasoning. Symbolic approaches represent knowledge through explicit formal structures, whereas neural approaches encode knowledge implicitly in distributed numerical parameters learned from data.

Ontologies provide a formal, machine-interpretable mechanism for representing domain meaning, enabling shared conceptualization, explicit definitions, and logical inference across heterogeneous data sources. Studies in semantic information systems show that ontology-based integration improves consistency and interpretability when combining multiple repositories (Noy and McGuinness, 2001), while ontology-driven reasoning supports the derivation of implicit knowledge through classification, validation, and constraint checking (Horrocks et al., 2005). Within LLM-based architectures, ontologies can serve as a semantic infrastructure layer that mediates between natural language queries and computational reasoning, allowing systems to map queries to formal concepts and produce answers grounded not only in retrieved text but also in explicit conceptual semantics.

Integration Patterns for Semantic Components Semantic-oriented artifacts can be integrated into QA architectures through complementary patterns such as semantic grounding modules that map natural language to formal representations, validation layers that check answers against domain constraints, inference augmentation that combines symbolic reasoning with neural generation, and explanation mechanisms that produce structured justifications. Hybrid architectures that integrate neural and symbolic components have shown strong potential in tasks requiring compositional generalization and formal inference, supporting the feasibility of these approaches.

Benefits for Decision Support Incorporating semantic-oriented components offers key advantages for decision-support scenarios, including improved conceptual validity through evaluation against formal definitions, greater consistency across queries, enhanced traceability via inspectable reasoning steps, better generalization through concept-based reasoning, and increased robustness to phrasing variations or retrieval noise. These properties are especially important in enterprise contexts, where decisions must be explainable, auditable, and aligned with domain policies.

5 Conclusions and Future Research Directions

This survey examined LLM-based question answering architectures from a structural and representational perspective, focusing on how design choices affect grounding, reasoning, and decision-oriented queries. By organizing approaches into a taxonomy spanning retrieval-based, graph-based, tool-augmented, agentic, and deliberative paradigms, we identified a common pattern: while current systems are effective at accessing and aggregating information, they largely treat knowledge as retrievable content, limiting their ability to support conceptual interpretation or derive implicit knowledge when it is not explicitly present in data sources.

To address this limitation, we discussed semantic-oriented extensions as complementary components that introduce explicit conceptual models, symbolic reasoning, and formal domain structures, enhancing capabilities beyond retrieval and generation. Advancing QA toward robust decision support requires a shift from information access to semantic interpretability and conceptual reasoning, raising challenges such as scalable semantic modeling, integration of symbolic and neural methods, and new evaluation frameworks. Future research lies at the intersection of NLP, knowledge representation, and decision-support systems, with directions including ontology induction, adaptive grounding, hybrid inference, and concept-aware interactive systems.

REFERENCES

Asai, A., Wu, Z., Wang, Y., Sil, A., and Hajishirzi, H. (2024). Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*.

Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? *FAccT*.

Berners-Lee, T., Hendler, J., and Lassila, O. (2001). The semantic web. *Scientific American*, 284(5):34–43.

Brachman, R. J. and Levesque, H. J. (2004). *Knowledge Representation and Reasoning*. Morgan Kaufmann.

Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., and Larson, J. (2024). From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Gao, D., Wang, H., Li, Y., Sun, X., Qian, Y., Ding, B., and Zhou, J. (2024). Text-to-SQL empowered by large language models: A benchmark evaluation. *Proceedings of the VLDB Endowment (PVLDB)*, 17(5):1132–1145.

Gao, L., Ma, X., Lin, J., and Callan, J. (2023a). Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*.

Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., and Wang, H. (2023b). Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.

Gruber, T. R. (1993). A translation approach to portable ontology specifications. *Knowledge Acquisition*, 5(2):199–220.

Horrocks, I., Patel-Schneider, P., and van Harmelen, F. (2005). From shiq and rdf to owl: The making of a web ontology language. *Journal of Web Semantics*.

Karpukhin, V., Oguz, B., Min, S., Lewis, P., et al. (2020). Dense passage retrieval for open-domain question answering. In *EMNLP*.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Lipton, Z. (2018). The mythos of model interpretability. *Communications of the ACM*.

Ma, C., Chen, Y., Wu, T., Khan, A., and Wang, H. (2025). Large language models meet knowledge graphs for question answering: Synthesis and opportunities. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 24578–24597. Association for Computational Linguistics.

Noy, N. and McGuinness, D. (2001). Ontology development 101: A guide to creating your first ontology. *Stanford Knowledge Systems Laboratory Technical Report*.

Peng, B. et al. (2024). Graph retrieval-augmented generation: A survey. *ACM Computing Surveys*.

Plaat, A. et al. (2025). Agentic large language models: A survey. *arXiv preprint arXiv:2503.23037*.

Power, D. J. (2002). *Decision Support Systems: Concepts and Resources for Managers*. Quorum Books.

Sarthi, P., Abdullah, S., Tuli, A., Khanna, S., Goldie, A., and Manning, C. D. (2024). RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *International Conference on Learning Representations (ICLR)*.

Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., and Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Seabra, A., Cavalcante, C., Medeiros, G., Nepomuceno, J., Ruberg, N., Sallenave, V., and Lifschitz, S. (2025). Business intelligence for contract management using large language models and ai agents. In *International Conference on Smart Business Technologies*, pages 160–182. Springer.

Shuster, K. et al. (2021). Retrieval augmentation reduces hallucination in conversation. *Findings of EMNLP*.

Simon, H. A. (1960). *The New Science of Management Decision*. Harper & Brothers.

Sun, H., Bedrax-Weiss, T., and Cohen, W. (2019). Pullnet: Open domain question answering with iterative retrieval on knowledge bases and text. In *EMNLP*.

Taylor, J. et al. (2022). Hybrid hierarchical retrieval for open-domain qa. *Transactions of the ACL*.

Uschold, M. and Gruninger, M. (1996). Ontologies: Principles, methods and applications. *Knowledge Engineering Review*, 11(2):93–136.

Wei, J. et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*.

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., and Narasimhan, K. (2023). Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.